

Web Component Development in Java (CGI Programming)

Theory:

Introduction to CGI (Common Gateway Interface):

CGI (Common Gateway Interface) is a standard method used to generate dynamic web content. It allows a web server to interact with external programs written in languages like C, C++, or Java.

When a user sends a request through a web browser, the web server executes a CGI program and sends the output back to the browser as a response.

Process of CGI Programming:

1. The user sends a request from the browser.
2. The web server receives the request.
3. The server calls the CGI program.
4. The CGI program processes the request (may interact with a database).
5. The program generates output (HTML).
6. The server sends the response back to the browser.

Advantages of CGI:

- Simple and easy to understand
- Language independent (can use C, Java, Python, etc.)
- Works with almost all web servers
- Useful for small applications

Disadvantages of CGI:

- Slow performance (new process created for each request)
- High memory usage
- Not suitable for large-scale applications
- Difficult to maintain for complex systems